AS · Unit 2 · WPH12 Waves & Electricity

Superposition & Stationary Waves

What happens when two waves meet? They simply add up, point by point. That one idea, superposition, explains the loud and quiet spots between two loudspeakers, the bright and dark bands in a ripple tank, and the stationary waves on a guitar string that decide which notes it can play. We'll build it all up from the idea of phase.

SPEC 39 Wavefronts, coherence, path difference, superposition, interference and phase

WHAT THIS SPEC POINT ASKS

Know what wavefront, coherence, path difference, superposition, interference and phase mean.

The definitions

Term	Meaning
Phase	where a point is in its cycle of oscillation. **Phase difference** is how far one oscillation is ahead of or behind another, as a fraction of a cycle, given as an **angle** (degrees or radians)
Wavefront	a line (or surface) joining points on a wave that are **in phase**, e.g. along one crest
Path difference	the **difference in distance** travelled by two waves from their sources to the same point, in metres (or in wavelengths)
Superposition	when two or more waves meet, the resultant displacement at any point is the **sum of the individual displacements**
Coherence	two sources are coherent if they have the **same frequency** and a **constant phase difference**
Interference	the superposition of waves from **coherent** sources, giving a steady pattern of maxima and minima

Constructive and destructive interference

	Constructive	Destructive
Waves arrive	**in phase**	in **antiphase**
Phase difference	0, 360° (2π rad), …	180° (π rad), 540°, …
Path difference	**nλ** (n = 0, 1, 2, … a whole number)	**(n + ½)λ**: λ/2, 3λ/2, …
Result	larger amplitude: loud, bright	smaller amplitude: quiet, dark. Zero only if the two amplitudes are equal

Why coherence matters

- If the phase difference between two sources keeps changing, the maxima and minima move around faster than you can see or hear, so there is no steady pattern.
- **Two separate lamps** are not coherent: each atom emits light in short random bursts, so the phase relationship changes all the time (and the light contains many frequencies).
- **Coherent set-ups:** two loudspeakers driven by the **same** signal generator; one laser beam split by two slits or a grating.
- A laser is monochromatic (one frequency) and coherent.

How to write an interference explanation

1. Say the waves **superpose / interfere** where they meet.
2. Give the **path difference** at that point (nλ or (n + ½)λ, saying n is an integer).
3. Link it to **phase**: in phase or antiphase.
4. Name the result: constructive or destructive, so a maximum (loud/bright) or minimum (quiet/dark).

WORKED EXAMPLE

Two loudspeakers A and B are connected to the same signal generator and emit sound in phase at 850 Hz. A microphone is 2.60 m from A and 3.20 m from B. The speed of sound is 340 m s^{-1}.
(a) Show that the microphone is at a minimum.
(b) The frequency is increased slowly. Find the next frequency at which the microphone detects a maximum.
(c) Suggest why the sound at the minimum does not drop to zero.

Show solution

(a) Wavelength: $\lambda = v/f = 340/850 = 0.400$ m. Path difference: $3.20 - 2.60 = 0.60$ m. In wavelengths:

$$\frac{\Delta x}{\lambda} = \frac{0.60}{0.400} = 1.5$$

A path difference of 1.5λ means the waves arrive in antiphase ($\phi = 3\pi$), so there is destructive interference: a minimum.

(b) A maximum needs $\Delta x = n\lambda$, so $\lambda = 0.60/n$. Increasing f makes λ smaller than 0.400 m. The next allowed value is $n = 2$:

$$\lambda = \frac{0.60}{2} = 0.30 \text{ m} \quad \Rightarrow \quad f = \frac{v}{\lambda} = \frac{340}{0.30} = 1130 \text{ Hz } (3 \text{ s.f.})$$

(c) The microphone is further from B, so the wave from B has spread out more and has a smaller amplitude than the wave from A. Unequal amplitudes can't cancel completely. Sound reflected off the walls and floor also reaches the microphone.

MARK-SCHEME WORDING

- Coherent: "same frequency **and** constant phase difference / relationship". "Same phase" doesn't score (WPH12 Jan 2024 examiner report).
- Wavefront: "a line on which all points are in phase" (WPH12 Oct 2023).
- "Path difference = nλ **where n is an integer**": just "nλ" lost the mark for most candidates (WPH12 Oct 2024).
- "Phase difference must refer to an angle, path difference to a distance." "Out of phase" is not accepted for antiphase.

Often asked as: definitions (1 mark each), why two lamps aren't coherent, 6-mark interference explanations (two speakers, noise-cancelling headphones, beats).

COMMON MISTAKE

Mixing up phase and path difference: "a phase difference of λ/2" is wrong (it is π rad); "a path difference of 180°" is wrong (it is λ/2).

At the central point between two in-phase sources, the path difference is **zero**: don't write nλ or (n + ½)λ there.

Writing a list of facts in a 6-mark answer. Link each step with "so" or "because".

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Noise-cancelling headphones**: the cancelling wave must arrive in antiphase, giving destructive interference (WPH12 Oct 2020).
- **Beats**: two notes of 880 Hz and 882 Hz are not coherent, so their phase difference drifts; the sound is loud when they are in phase and quiet in antiphase (WPH12 Oct 2022). Answer in terms of phase, not path difference.
- **Thin film**: light reflected from the top and bottom of a coating of thickness d has path difference 2d. Use the wavelength **inside** the coating, λ_air ÷ n (WPH12 Oct 2023).

SPEC 40 Phase difference and path difference

WHAT THIS SPEC POINT ASKS

Convert between the phase difference and the path difference of two waves.

The relationship

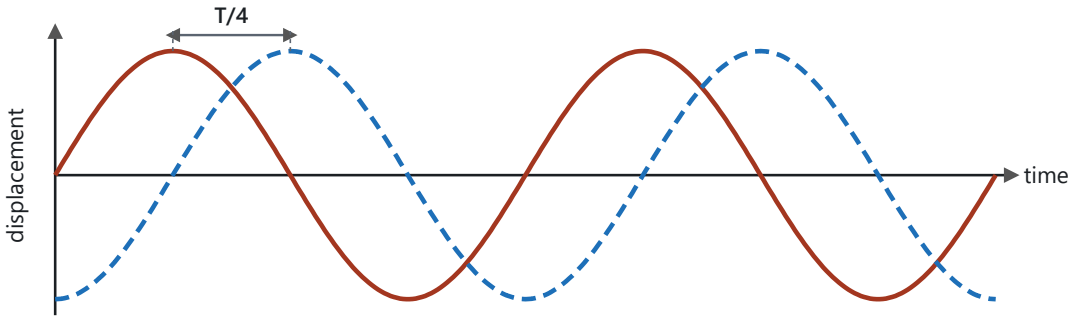
One whole wavelength of path difference = one whole cycle = 360° = 2π rad. So:

Phase difference from path difference

$$\phi = \frac{2\pi\,\Delta x}{\lambda} \qquad \text{or} \qquad \phi = 360^\circ \times \frac{\Delta x}{\lambda}$$

ϕ: phase difference / rad (or °) Δx: separation along the wave, or path difference / m
λ: wavelength / m

The same idea works for two points on one progressive wave a distance Δx apart (use Δx as the "path difference"), and for time: phase difference = (Δt / T) × 360°.



Solid A, dashed B. B lags A by a quarter cycle: phase difference = 90° = π/2 rad

Two oscillations a quarter of a period (or quarter of a wavelength) apart have a phase difference of 90°, π/2 rad.

Path difference	Phase difference	Interference
0, λ, 2λ	0 (or 360°, 720°)	constructive
λ/4	90° = π/2 rad	neither
λ/2, 3λ/2	180° = π rad	destructive
3λ/4	270° = 3π/2 rad	neither

WORKED EXAMPLE

two loudspeakers

Two loudspeakers driven by the same 500 Hz signal. Speed of sound 340 m s^{-1}. A point P is 4.10 m from one speaker and 3.08 m from the other.

λ = v/f = 340 ÷ 500 = 0.680 m.

Path difference = 4.10 − 3.08 = 1.02 m = 1.02 ÷ 0.680 = 1.5 λ.

Phase difference = 1.5 × 360° = 540°, which is the same as **180° (π rad)**: antiphase.

So destructive interference: P is a **quiet** point (silent only if the two amplitudes are equal).

Two points on one of these waves 0.17 m apart: 0.17 ÷ 0.680 = 0.25 λ, so phase difference = **90° (π/2 rad)**.

Relating phase difference to path difference

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 One wavelength along a progressive wave corresponds to one complete cycle of phase:

$$\Delta x = \lambda \quad \longleftrightarrow \quad \phi = 2\pi \text{ rad}$$

2 A distance Δx is the fraction $\Delta x/\lambda$ of a wavelength, so it gives the same fraction of a full cycle:

$$\frac{\phi}{2\pi} = \frac{\Delta x}{\lambda}$$

3 Multiply both sides by 2π:

$$\boxed{\phi = \frac{2\pi\,\Delta x}{\lambda}}$$

Check the special cases: $\Delta x = \lambda/2$ gives $\phi = \pi$ (antiphase); $\Delta x = \lambda$ gives 2π (back in phase).

WORKED EXAMPLE

A progressive wave has frequency 25 Hz and speed 15 m s^{-1}. Points P and Q lie 0.45 m apart along the direction the wave travels.
(a) Calculate the phase difference between P and Q.
(b) State the smallest distance from P to a point that oscillates in antiphase with P.

Show solution

(a) First find the wavelength from $v = f\lambda$:

$$\lambda = \frac{v}{f} = \frac{15}{25} = 0.60 \text{ m}$$

Then the phase difference:

$$\phi = \frac{2\pi\,\Delta x}{\lambda} = \frac{2\pi \times 0.45}{0.60} = 1.5\pi \text{ rad} = \frac{3\pi}{2} \text{ rad } (270^\circ)$$

Q is further along, so Q lags P by $3\pi/2$ rad. (You could equally say Q leads P by $\pi/2$ rad: lagging by three quarters of a cycle looks the same as leading by one quarter.)

(b) Antiphase means $\phi = \pi$, which is half a wavelength: $\lambda/2 = 0.30$ m.

MARK-SCHEME WORDING

- "Phase difference = 2π × (path difference)/λ" or the degree version, with the answer as an **angle**.
- Whole number of wavelengths → in phase; odd number of half wavelengths → antiphase.

Often asked as: one-mark MCQs converting λ fractions to angles (3λ/8 = 135°; λ/8 = π/4; 7λ/4 = 3.5π rad, the same as 1.5π rad).

COMMON MISTAKE

Forgetting to subtract whole cycles: 540° and 180° describe the same phase relationship.

Mixing degrees and radians in one line. Pick one.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

One mark scheme (WPH12 Oct 2019, MCQ) wrote that 7λ/4 is "3.5 radians". It is **3.5π rad** (equivalent to 1.5π rad). The answer letter was right; the working was not. Don't copy it.

SPEC 41 Stationary (standing) waves

WHAT THIS SPEC POINT ASKS

Know what a stationary wave is and how one forms, and identify its nodes and antinodes.

How one forms

1. A wave travels along a string (or through air in a pipe) and is **reflected** at the end (the fixed end, the bridge, the closed end of a pipe, a metal plate…).
2. The incident and reflected waves have the **same frequency** (and similar amplitude) and travel in **opposite directions**.
3. They **superpose**. At some points they always cancel (**nodes**); at others they always reinforce (**antinodes**).
4. A steady pattern only appears at certain frequencies (the resonant frequencies), when a whole number of half-wavelengths fits the length.

Nodes and antinodes

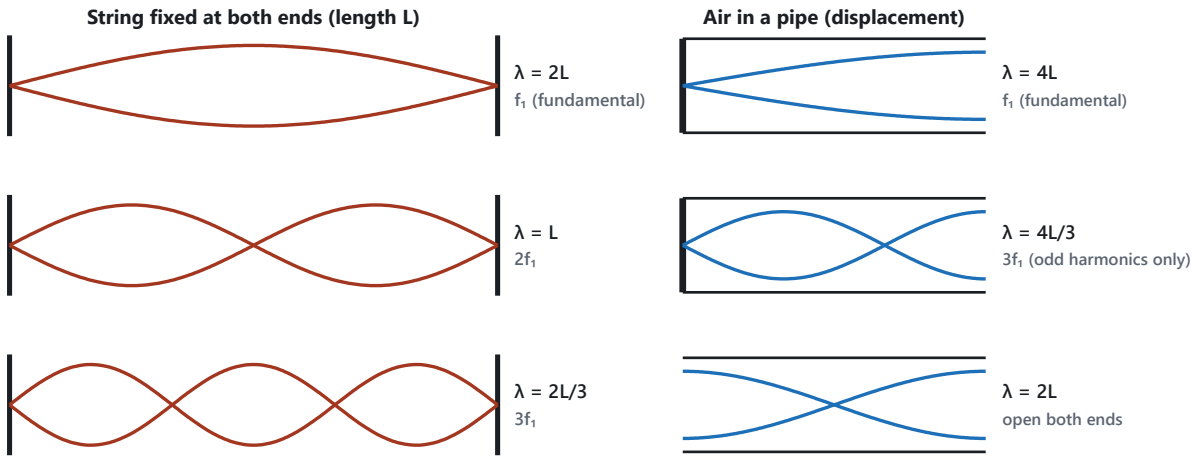
- **Node:** a point of **zero amplitude**. The two waves always arrive in antiphase: **destructive** interference.
- **Antinode:** a point of **maximum amplitude**. The two waves always arrive in phase: **constructive** interference. The displacement at an antinode still varies, from +A through zero to −A.

- Node to next node = **λ/2**. Node to next antinode = **λ/4**.

Stationary v progressive

	Progressive wave	Stationary wave
Energy	transferred along the wave	**not transferred** (stored in the oscillation)
Amplitude	the same at every point (no losses)	varies with position: zero at nodes, max at antinodes
Phase	changes steadily along the wave over one λ	all points between adjacent nodes in phase; adjacent loops in antiphase
Wavelength	distance between adjacent points in phase	twice the node-to-node distance

Harmonics on strings and in pipes



Closed end = node. Open end = antinode. Fixed string end = node.

A string fixed at both ends has a node at each end. A pipe closed at one end has a node at the closed end and an antinode at the open end. A pipe open at both ends has antinodes at both ends.

System	Fundamental (first harmonic)	Higher harmonics
String fixed at both ends, length L	λ = 2L	λ = 2L/n, f = n f_1 (n = 1, 2, 3, …)
Pipe closed at one end	λ = 4L	odd only: $3f_1$, $5f_1$, …
Pipe open at both ends	λ = 2L	all: $2f_1$, $3f_1$, …

Measuring λ with microwaves

Point a microwave transmitter at a metal plate. Move a detector along the line between them: the reading rises and falls as it passes antinodes and nodes. Measure across several nodes: node-to-node distance = λ/2.

WORKED EXAMPLE

wavelength from nodes

A microwave detector finds 6 nodes spread over 7.0 cm (that is 5 node-to-node gaps).

λ/2 = 7.0 ÷ 5 = 1.4 cm, so λ = 2.8 cm = 0.028 m.

f = c/λ = 3.00 × 10^8 ÷ 0.028 = **1.1 × 10^{10} Hz** (2 s.f.).

A string 1.70 m long showing 4 loops: 4 × λ/2 = 1.70 m, so λ = **0.85 m** (WPH12 Jan 2020 style).

WORKED EXAMPLE

A microwave transmitter faces a flat metal plate. A detector moved along the line between them finds a series of maxima and minima. Moving the detector from one minimum to the sixth minimum after it covers a distance of 8.4 cm.
(a) Explain why the minima occur.
(b) Calculate the frequency of the microwaves.
(c) Explain why the signal at each minimum is small but not zero.

Show solution

(a) Microwaves reflect from the plate and travel back towards the transmitter. The incident and reflected waves have the same frequency and travel in opposite directions, so they superpose to form a stationary wave. The minima are the nodes, where the two waves are always in antiphase and cancel.

(b) The starting minimum and the sixth one after it are 6 node-to-node gaps apart, and each gap is $\lambda/2$:

$$6 \times \frac{\lambda}{2} = 8.4 \text{ cm} \quad \Rightarrow \quad \lambda = 2.8 \text{ cm} = 0.028 \text{ m}$$

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8}{0.028} = 1.1 \times 10^{10} \text{ Hz (2 s.f.)}$$

(c) The reflected wave has a smaller amplitude than the incident wave (some energy is absorbed and it spreads out), so the two cannot cancel completely at the nodes.

MARK-SCHEME WORDING

- Formation: "wave **reflected** at [the named reflector]", "two waves travelling in **opposite directions**" (or "reflected back on itself"), "same frequency", "**superposition / interference**".
- Node: "zero / minimum amplitude", "destructive interference". Antinode: "maximum amplitude", "constructive interference".
- "Node to node = λ/2."

Often asked as: explain how the stationary wave forms in a context (string, tube above water, bridge cable, metal plate), distinguish nodes and antinodes, λ from a pattern, tube resonance frequencies.

COMMON MISTAKE

Quoting a textbook definition without the context. The first mark often needs the actual reflector, e.g. "reflected by the water surface"; very few candidates said this (WPH12 Oct 2021 examiner report).

"Two opposite waves". Say "travelling in opposite directions".

Giving λ as twice the string length when there are several loops (WPH12 Jan 2020 examiner report). Count the loops first.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Closed tube resonance**: a tube above water, or closed at one end, resonates first when its length is λ/4, so λ = 4L. E.g. 440 Hz × (4 × 0.193 m) = 340 m s^{-1} (WPH12 Oct 2021).
- **End correction**: the antinode sits slightly beyond the open end of a tube, so the real λ is a bit longer than 4L and a speed worked out from 4L comes out too low (WPH12 Jun 2025).
- To lower the pitch of a recorder-type instrument (open pipe, λ = 2L), make the air column longer (WPH12 Jan 2021).

SPEC 42 Speed of a transverse wave on a string

WHAT THIS SPEC POINT ASKS

Use $v = \sqrt{(T/\mu)}$ for the speed of a transverse wave on a stretched string.

The equation

Speed of a transverse wave on a string

FORMULA SHEET

$$v = \sqrt{\frac{T}{\mu}}$$

v: wave speed / m s⁻¹ T: tension / N μ: mass per unit length / kg m⁻¹

- v = wave speed on the string (m s⁻¹); T = **tension** (N); μ = **mass per unit length** (kg m⁻¹) = mass of the string ÷ its length. It is on the formulae sheet.
- Tighter string → faster wave. Heavier string → slower wave.
- If the tension comes from a hanging mass m: **T = mg**, with g = 9.81 N kg⁻¹.

Linking it to frequency

For the fundamental of a string fixed at both ends, λ = 2L (L = **vibrating** length, between the two fixed points). Using v = fλ:

Fundamental frequency of a string

$$f_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$$

f_1: fundamental (first harmonic) frequency / Hz L: length of the vibrating string / m
T: tension / N μ: mass per unit length / kg m⁻¹

Change	Effect on f_1
double the length L	f halves (f ∝ 1/L)
quadruple the tension T	f doubles (f ∝ √T)
quadruple μ (heavier string)	f halves (f ∝ 1/√μ)

WORKED EXAMPLE

frequency of a loaded string

A string with μ = 1.20 × 10⁻³ kg m⁻¹ passes over a pulley and holds a 0.500 kg mass. The vibrating length is 0.800 m.

T = mg = 0.500 × 9.81 = 4.905 N.

v = √(T/μ) = √(4.905 ÷ 1.20 × 10⁻³) = √4088 = 63.9 m s⁻¹.

Fundamental: λ = 2L = 1.60 m, so f = v/λ = 63.9 ÷ 1.60 = **40.0 Hz** (3 s.f.).

WORKED EXAMPLE

tuning by tension

A string sounds 432 Hz. What percentage increase in tension raises it to 440 Hz?

f ∝ √T, so T ∝ f². New T ÷ old T = (440 ÷ 432)² = 1.037. The tension must rise by about **3.7 %**.

Frequencies of the harmonics on a string

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 Both ends are nodes, and each loop is half a wavelength. With n loops on a string of length L:

$$L = n\frac{\lambda}{2} \quad \Rightarrow \quad \lambda = \frac{2L}{n}$$

2 The wave speed on the string is fixed by T and μ, so use $v = f\lambda$ to find the frequency:

$$f = \frac{v}{\lambda} = \frac{nv}{2L}$$

3 Substitute $v = \sqrt{T/\mu}$:

$$\boxed{f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}}$$

4 For the fundamental, $n = 1$:

$$f_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$$

So $f_1 \propto 1/L$, $f_1 \propto \sqrt{T}$ and $f_1 \propto 1/\sqrt{\mu}$: exactly the three relationships Core Practical 5 tests. Short, tight, thin strings give high notes.

WORKED EXAMPLE

A guitar string has a vibrating length of 0.648 m and a mass per unit length of 1.15×10^{-3} kg m^{-1}. It plays a fundamental frequency of 110 Hz.
(a) Calculate the speed of waves on the string.
(b) Calculate the tension in the string.
(c) The player presses the string against a fret so that the vibrating length becomes 0.432 m. Calculate the new fundamental frequency.
(d) The tension is then increased by 10%. By what factor does the frequency change?

Show solution

(a) For the fundamental, $\lambda = 2L = 2 \times 0.648 = 1.296$ m:

$$v = f\lambda = 110 \times 1.296 = 142.6 \text{ m s}^{-1} \approx 143 \text{ m s}^{-1}$$

(b) Square $v = \sqrt{T/\mu}$ and rearrange:

$$T = \mu v^2 = 1.15 \times 10^{-3} \times 142.6^2 = 23.4 \text{ N (3 s.f.)}$$

(c) T and μ are unchanged, so v is unchanged and $f_1 \propto 1/L$:

$$f_1 = \frac{v}{2L} = \frac{142.6}{2 \times 0.432} = 165 \text{ Hz (3 s.f.)}$$

(d) $f \propto \sqrt{T}$, so the frequency changes by a factor of $\sqrt{1.10} = 1.049$: it rises by about 4.9%.

MARK-SCHEME WORDING

- "Use of T = mg", "use of v = √(T/μ)", "recognises λ = 2L", then "use of v = fλ".
- Ratio questions: "f ∝ √T" so "T ∝ f²".

Often asked as: speed and frequency of a loaded string, choose which string gives a required note, small tension changes, MCQs on what changes v.

COMMON MISTAKE

Multiplying μ by the length. μ is already "per metre" (WPH12 Jan 2020 examiner report). If you're given a total mass, μ = m ÷ L.

Using g = 10 N kg^{-1} costs a mark. Use 9.81.

Using the full string length when only part of it vibrates (bridge to bridge).

No unit on the final answer.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Bridge cables: μ = cable mass ÷ cable length (e.g. 2.10 × 10^3 kg ÷ 87.0 m = 24.1 kg m^{-1}), and the tension is the load shared between the cables. You then show which frequencies the cable can support (WPH12 Oct 2024).

SPEC 43 Core practical 5: frequency of a vibrating string

WHAT THIS SPEC POINT ASKS

Investigate how the length, tension and mass per unit length of a string affect its frequency of vibration.

CORE PRACTICAL

Core practical 5: method in brief

1. Fix one end of a string to a **vibration generator** driven by a **signal generator**. Run the string over a bridge and a pulley, with a mass hanger on the end (tension T = mg).
2. Set a vibrating length L between the vibration generator (or a bridge) and the pulley bridge. Measure it with a metre rule.
3. Slowly increase the frequency from low until the string shows **one loop of maximum amplitude** (the fundamental). Record f from the signal generator.
4. **Length:** move the bridge to change L; keep T and μ fixed.
5. **Tension:** add masses; keep L and μ fixed.
6. **Mass per unit length:** use different strings; find each μ by weighing a measured long length (mass ÷ length); keep L and T fixed.

7. Repeat each reading (approach the resonance from above and below) and average.

Turning the results into straight lines

From f = (1/2L)√(T/μ):

Investigation	Plot	Straight line through origin, gradient
Length	f against 1/L	½√(T/μ)
Tension	f² against T	1/(4L²μ)
Mass per unit length	f against 1/√μ	(1/2L)√T

Errors and safety

- Judging the exact point of maximum amplitude is hard. Approach it from both sides and average; or measure across several loops at a higher harmonic.
- **Parallax** when reading L: hold the ruler close to the string and read at eye level.
- The generator dial may be inaccurate: check f with an oscilloscope (measure T, f = 1/T).
- Safety: masses can fall. Put a cushion or box underneath and keep feet clear. A snapping string can whip: wear eye protection.

WORKED EXAMPLE

μ from an f² against T graph

With L = 0.600 m, a graph of f² against T has gradient 1.80 × 10³ Hz² N⁻¹.

Gradient = 1/(4L²μ), so μ = 1 ÷ (4 × 0.600² × 1.80 × 10³) = 1 ÷ 2592 = **3.86 × 10⁻⁴ kg m⁻¹** (3 s.f.).

WORKED EXAMPLE

In Core Practical 5 a student keeps the tension at 4.9 N and plots f_1 against $1/L$. The best-fit line passes through the origin with gradient 47.5 Hz m. Calculate the mass per unit length of the wire.

Show solution

Since $f_1 = \dfrac{v}{2} \times \dfrac{1}{L}$, the gradient is $v/2$:

$$v = 2 \times 47.5 = 95.0 \text{ m s}^{-1}$$

Then from $v = \sqrt{T/\mu}$:

$$\mu = \frac{T}{v^2} = \frac{4.9}{95.0^2} = 5.4 \times 10^{-4}\ \text{kg m}^{-1}\ (2\ \text{s.f.})$$

MARK-SCHEME WORDING

- "Show that $f^2 \propto W$" (5 marks): equate $v = f\lambda$ and $v = \sqrt{(T/\mu)}$; rearrange to $f^2 = T/(\mu\lambda^2)$ or $T/(4L^2\mu)$; $T = W$; "μ and λ (or L) are constant"; so $f^2 \propto W$, "y = mx with no c / y-intercept is 0" (WPH12 Oct 2020).
- Measuring f: "read from the signal generator" or "use an oscilloscope: measure T, f = 1/T".
- Uncertainty: "difficult to judge the exact position of the node"; "ruler not close to the string, so parallax" (WPH12 Jan 2022).

Often asked as: show a graph is a straight line through the origin, find μ or a mass from a gradient, how to measure f, sources of uncertainty, MCQ on which variable is not a control.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Turning an unfamiliar equation with two squared quantities into y = mx + c and reading μ from the gradient, e.g. f^2 against T with gradient 532 $\text{Hz}^2\ \text{N}^{-1}$ giving $\mu = 6.5 \times 10^{-4}\ \text{kg m}^{-1}$ (WPH12 Jan 2025).

Quick checks

QUICK CHECK

Two points on a progressive wave are 0.15 m apart along the direction of travel. The wavelength is 0.60 m. What is the phase difference between them?

- 0.25 rad
- $\pi/4$ rad
- $\pi/2$ rad
- π rad

The points are $0.15/0.60 = 0.25$ of a wavelength apart, so $\phi = 0.25 \times 2\pi = \pi/2$ rad. "0.25 rad" forgets to multiply the fraction of a cycle by 2π.

QUICK CHECK

Which of these is needed for two sources to be coherent?

- The same amplitude and the same frequency
- The same frequency and a constant phase difference
- The same amplitude and zero phase difference
- The same wavelength and the same amplitude

Coherence is about a steady phase relationship, which needs equal frequencies. The amplitudes can differ (you just won't get complete cancellation), and the phase difference can be any value as long as it doesn't change.

QUICK CHECK

Two loudspeakers are connected so that they vibrate in antiphase. A microphone is placed at a point exactly the same distance from both. What does it detect?

- A maximum, because the path difference is zero
- A minimum, because the waves arrive in antiphase
- Alternating maxima and minima
- No interference, because the sources are not coherent

With zero path difference the waves arrive with the same phase difference they left with: π. Antiphase sources are still coherent, because the phase difference is constant.

QUICK CHECK

Points X and Y lie between the same pair of adjacent nodes on a stationary wave. X is closer to the antinode. Which statement is correct?

- X and Y have the same amplitude but are out of phase.
- X and Y are in antiphase.
- X and Y are in phase but X has a larger amplitude.
- X and Y have the same amplitude and are in phase.

Within one loop of a stationary wave every point moves up and down together (in phase), but the amplitude increases from zero at the nodes to a maximum at the antinode.

QUICK CHECK

On a stationary wave of wavelength λ, what is the distance between a node and the nearest antinode?

- λ/4
- λ/2
- λ
- 2λ

Adjacent nodes are $\lambda/2$ apart and the antinode lies halfway between them, so it is $\lambda/4$ from each node.

QUICK CHECK

The tension in a vibrating string is made four times bigger. The length and the string are unchanged. What happens to the fundamental frequency?

- It is quartered.
- It doubles.
- It is four times bigger.
- It stays the same, because the length is unchanged.

The wavelength of the fundamental is fixed at $2L$, but $v = \sqrt{T/\mu}$ doubles when T is multiplied by 4. Since $f = v/\lambda$, the frequency doubles too.